



Khalsa College of Engineering & Technology

ਖਾਲਸਾ ਕਾਲਜ ਆਫ ਇੰਜੀਨੀਅਰਿੰਗ ਐਂਡ ਟੈਕਨੋਲੋਜੀ

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RULES FOR TECH URJA 2K26

Rules for Paper Presentation

- A team should have a minimum of 2 and a maximum of 3 participants.
- The duration of the presentation should be between 4 and 6 minutes.
- The number of slides to be prepared should not be less than 8 and not more than 12.
- Each team will be asked two questions by judges related to presentation.
- The presentation will be judged by judges on the basis of : **Content of presentation, Confidence and verbal Skills, Use of graphics, Sound, video and question round**

Topics of Paper Presentation:

- Robotics & Automation
- Automatic Innovations
- Neural Network
- Artificial Intelligence
- Generative AI
- Explainable deep learning models for high-risk decision-making.
- ECG & ECHO
- Introduction to Histotechnology & Its Core Techniques.
- AI for Climate Change Prediction

Topics of UrjaXPO (Project Display)

Paramedical

- Brain Computer Interface
- Deep-fake detection software
- Radiological Clinical Positioning



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S. No.	Name of Event	Rules
1	Bridge Building	3 members make a team. - Only wooden popsicle sticks and white glue can be used to make your bridge. - You may use no more than 120 popsicle sticks. - The bridge must be at least 300 mm long and have an overall width of at least 100 mm. - Please remember that load testing may result in the destruction of the bridge. - Total time allotted for the event is 60 minutes, and after 2 hours of making load testing will be done in front of judges. - The load will be applied in the center of the span on the roadbed surface. - A bridge may be considered to have "failed" under the test load if the bridge components break, crack, or separate, causing the collapse. - Items used: Popsicle sticks, gum, and construction paper.
2	Solid PRO Test	- Maximum one participant per college is allowed. - The topic will be announced on the spot.
3	Object Scaling Drawing	- Maximum one participant per college is allowed. - A problem to test the logic will be given on the spot. - Participants must bring their own required items
4	CATAPULT	2 members make a team. - Items used: Popsicle sticks, gum, construction paper, cardboard, and any waste material. - Total time allotted for the event is 60 minutes after 2 hours of making CATAPULT. - Testing will be done by calculating the distance of the object through it.
5	AUTOCAD 2D PLAN	- A drawing will be given to participants, & they will have to draw it in AUTOCAD. - The drawing should be made with proper dimensioning. - Any discrepancy leads to the disqualification of the candidate. - Total time allotted for the event is 25 minutes and 5 minutes to think over the matter. - Candidates who finish the task first will win the first round. - The shortlisted candidate will appear in round 2. - The same rules must be followed in round 2

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6	SURVEY HUNT	• A team shall be of 3 members only. • In the first round, a screening test shall be conducted, and the top 4 teams will be selected for the second round based on the accuracy of the work. • Time allotted for the task is 1 hour.
	Tech Whiz	• One participant per team is allowed. • Logos and computer parts will be shown on the spot. • A total of 2 wrong answers out of 5 will result in disqualification.
7	Logic X (Programming in C & C++)	• A maximum of two participants per team is allowed. • A problem to test the logic will be given on the spot. • C/C++ language will be used to solve the given problem • Judgment of this event will be based on solution strategy and time consumption used to solve the problem, correctness, and completeness of the program.
9	Logic Gates	• One participant per team is allowed. • The statement problem will be provided on the spot. • The logic will have to be built on the basis of proper examination of the given result provided with the Logic Gates to be used
10	Short Movie (6-8 Minutes)	• Max. 8 teams • 10 students in each team • The movie should be on the topic given. • The title of the movie is "Visa Lag Gaya" and you must create a movie on this topic. • No abusiveness or vulgarity will be allowed; otherwise, they will be disqualified. • The movie will be submitted to the coordinators on the said date. • The short movie must be submitted by 7th April. • We will add you to a WhatsApp group after registration, in which you will have to submit a short movie.



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11.	On the Spot Poster Making	• The theme for the poster should be related to AI, marketing & business. • Each Participant can post only one poster. • The poster must be made on the A3-size (11.7 * 16.5 inches) sheet with margins (0.5 inches) drawn with a black pen/marker. • A poster can be made using pencil, crayon, watercolor, oil paint, etc. • Photographs, wires, other 3D objects, and printouts won't be acceptable. It should be original and hand-made. • The topic of the poster and the theme should be clearly mentioned at the top margin of the poster. • The decision of the judges will be final
12.	Calligraphy	• Only cursive calligraphy is allowed. Mixing print and cursive styles is not permitted. • Writing must be done using blue or black ink only. • Participants must use the paper provided by the organizers. • The content/theme given by the organizers must be strictly followed. • Thin upstrokes and thick downstrokes should be clearly visible. • Letters must follow a straight baseline with a consistent slant. • Uniform spacing should be maintained between letters, words, and lines. • Capital letters must be neat and proportional to lowercase letters. • Minimal flourishes are allowed; excessive decoration will lead to marks deduction. • Overwriting, erasing, cutting, or using correction fluid is not allowed. • The sheet must be clean and free from ink spills or smudges. • Participants must complete the work within the allotted time. • The judges' decision will be final and binding.
13	Code Sprint (coding Challenge in JAVA & Python)	• Max 2 Participants • A problem to test logic will be given on the spot. • No built-ins to be used • Use of Internet is strictly prohibited
14	Bug Hunters (Debugging Contest)	• Only 1 Participant is allowed • Use of Internet is strictly prohibited • Pre-written code will be given • Only error correction is allowed



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15	Mapping Semester-Based Learning CSE Model	• Max 2 Participants • Group subjects by core computing concepts such as data computation, systems, intelligence, and security.
16	Circuit Design Challenge	• Participants design simple PCBs (e.g., LED flashers). • Judged on design time, optimization, and creativity. • LED Blink Challenge Task: Make an LED blink using given components. • Skills: Basic electronics • Time: 5–10 minutes • Twist: Blink in a given pattern (slow–fast–slow) • Each team will consist of two members.
17	E-puzzle	• Answer as many basic ECE questions as possible in 60 seconds. • Only 1 participant in a team is allowed.
18	Robo War	• Students design the robot and try to push it out from the marked area. • Each team will consist of two members.

19	Towel Art	• Entry per institution will be one. • A team of 2 students can participate at a time. • The time duration for this event is 15 mins. • Each team will bring their own towels for the event. • Mobile phones are not allowed. • It should be done in front of the judges.
20	No Flame Cooking	• Entry per institution will be one. • A team of 2 students can participate at a time. • The time duration for this event is 30 mins. • Ingredients should be brought by participants themselves. • Mobile phones are not allowed. • Event judges' decisions will be final and binding. • Preparation should be done in front of the judges.



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21	Preparation of Mocktails	• Entry per institution will be one. • Each team will consist of 2 members. • The time duration for this event is 30 mins. • Each team will bring their own ingredients for the event. • Mobile phones are not allowed. • Preparation should be done in front of the judges.
22	Rangoli	• Size: 2'x2' to 4'x4' (5% permissible variation) • Only one medium may be used: poster colors, flower petals, sawdust, pulses, or rice (without pasting). • The art form may be freehand, pictorial, or descriptive (e.g., mandna, alpna, alekhan, Kollam, and Rangoli). • Participants must prepare the Rangoli within the space provided by the organizers. • Topic/theme will be given on the spot by the judges. • Accessories (e.g., sieve, spoon, scale) are allowed.
23	Mehndi	• The event will be conducted on a subject/idea given on the spot. • Additional instructions will be announced on the spot. • Participants must bring their own materials; only traditional Indian motifs are allowed (no Arabian Mehndi or glitter). • Credit will be given for originality, creativity, and aesthetic sense. • No handprints, molds, or decorative materials are permitted. • Mehndi must extend to the palmar side of the forearm (minimum 6 inches).
24	Best out of waste	• Only waste or discarded materials must be used; new decorative items are not allowed. • The item must be made on the spot within the given time limit; pre-made items are prohibited. • Two members per team is allowed. • Creativity and originality are mandatory; copied designs may lead to disqualification. • Use of hazardous or unsafe materials is strictly prohibited. • The judges' decision will be final and binding.



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25	Solo Singing	• Time for Stage/Instrument Setting: Up to 1 minute • Maximum accompanists: 02 • Instrumental Music Is allowed for support. • Professional accompanists must sit/stand separately, not lead, and wear different attire from participants. • Language: English, Hindi, or Punjabi • Judgment based on singing quality, composition, rhythm, coordination, and general impression.
26	Duet/Group Singing	• Time for Stage/Instrument Setting: Up to 1 minute • Maximum accompanists: 02 • Instrumental Music Is allowed for support. • Professional accompanists must sit/stand separately, not lead, and wear different attire from participants. • Language: English, Hindi, or Punjabi • Judgment based on singing quality, composition, rhythm, coordination, and general impression.

27	Solo Dance (Boys & Girls)	• Time for Stage/Instrument Setting: Up to 2 minutes • Time starts with the first beat of the music. • Acrobatic actions are not allowed; double/triple standing is permitted. • Participants must remove their sets and properties immediately after the performance. • Judgment based on rhythm, formation, expression, costumes, makeup, overall effect, symmetry, composition, and general impression (e.g., grace and expressions)
28	Duet/Group Dance (Boys & Girls)	• Time for Stage/Instrument Setting: Up to 2 minutes • Time starts with the first beat of the music. • Acrobatic actions are not allowed; double/triple standing is permitted. • Teams must remove their sets and properties immediately after the performance. • Judgment based on rhythm, formation, expression, costumes, makeup, overall effect, symmetry, composition, and general impression (e.g., grace and expressions)



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29	DSLR Photography	• The team has a maximum of one participant • The student who clicks the BEST 10 photographs of the coverage of the tech fest will be the winner.																
29	eSports Competition	• Cheating, hacks, emulators, stream sniping, or teaming = instant disqualification. • Only mobile devices allowed—no PC/emulators; bring your own charged phone + ID proof. • Abusive chat, harassment, or damaging any setup = penalty or disqualification. • Arrive on time for room joining & setup; late teams forfeit the match. BGMI - Rules 1. General Rules • Each team must consist of 4 players (squad). • Players must use their own mobile devices (phones only). • An internet/data connection is the responsibility of the players. • Players must join the custom room lobby on time. Late entries will not be allowed. 2. Match Format • Mode: Squad (TPP) • Maps: Erangel + Miramar (Mandatory) • Total Rounds: 3 (2 Matches – Erangel, 1 Match – Miramar) • The final ranking will be based on total kills + placement points. 3. Scoring System <table><tr><th>Placement</th><th>Points</th></tr><tr><td>1st</td><td>10</td></tr><tr><td>2nd</td><td>6</td></tr><tr><td>3rd</td><td>5</td></tr><tr><td>4th</td><td>4</td></tr><tr><td>5th – 6th</td><td>3</td></tr><tr><td>7th – 8th</td><td>2</td></tr><tr><td>9th – 10th</td><td>1</td></tr></table>	Placement	Points	1st	10	2nd	6	3rd	5	4th	4	5th – 6th	3	7th – 8th	2	9th – 10th	1
Placement	Points																	
1st	10																	
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Per Kill

1

4. Fair Play Rules

- No cheating, hacking, or third-party applications allowed (immediate disqualification).
- No teaming with other squads outside your own team.
- No abusive language in the lobby or in-game chat.

5. Technical / Disconnection Rules

- If a player disconnects during the match, the game will continue.
- No rematch will be provided due to device, network, or power issues.

6. Additional Tournament Rules

- Tie-Breaker: If two teams have the same total points, the team with more total kills will rank higher.
- Teams may be asked to submit match result screenshots if required by the organizers.
- Players must follow all instructions given by the tournament officials during the event.
- The decision of the tournament organizers will be final in case of any dispute.

7. Prizes & Recognition

- Winner Team: Cash Prize
- MVP (Most Kills Overall): Cash Prize

Free Fire Max - Rules

1. General Rules

- Each team must consist of 4 players (squad).
- Minimum player level requirement: 30+.
- Players must have played at least 30 matches in the current BR season.
- Players must use their own mobile devices (phones only).
- An internet/data connection is the responsibility of the players.
- Players must join the custom room lobby on time. Late entries will not be allowed.

2. Match Format

- Mode: Squad (Battle Royale).
- Maps: Bermuda, Kalahari, Purgatory, Nexterra.
- Total Rounds: 4 (One match on each map).



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- The final ranking will be based on total kills + placement points.

3. Scoring System

Placement	Points
1st	10
2nd	6
3rd	5
4th	4
5th – 6th	3
7th	2
Per Kill	1

4. Fair Play Rules

- No cheating, hacking, or third-party applications allowed (immediate disqualification).
- No teaming with other squads outside your own team.
- No abusive language in the lobby or in-game chat.
- Gun skins must be turned OFF.

5. Technical / Equipment Rules

- Participants must bring their own equipment, including a mobile phone, internet access, and headset/earphones.
- Charging boards will not be provided, so keep your device fully charged before matches.
- Recording of gameplay must be saved.
- All maps must be completely downloaded before the event starts.

6. Additional Tournament Rules

- After completion of each match, screenshot submission is compulsory.
- All squad members must be present during the match.
- If two teams have the same total points, the team with more total kills will rank higher.
- The decision of the tournament organizers will be final in case of any dispute.



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30	UrjaXPO	• Every team has a maximum of 4 members. • The project should be related to science & technology based on either hardware or software. • The project must be working or be a demo model. • Copied projects from the internet without modification are not allowed. • Each team will bring their own poster prepared for the project idea. • Set-up time will be given to you before the exhibition starts. • Every team member must have ID proof. • Any damage to other projects will lead to disqualification.
31	Tech Reels	• Team has maximum two participants. • Theme/ Topic will be given on the spot • Time allowed 60 mins • Duration of each reel should be 45–60 seconds. • No abusiveness, vulgarity or copyrighted material; otherwise, disqualification. • Judgment will be based on creativity, content relevance, editing quality, engagement and technical aspects. • The decision of the Judges will be final and binding.
32	QuizPedia	• Each team shall have three members only. • The quiz will be conducted in two rounds. a) Screening Round : • All the participants can participate in the screening round, • from which 4 teams will be selected and those 4 teams will participate in the final rounds of the quiz. • The 4 teams selected will be from different college/institute. b) Final Rounds: Final Round will consist of following sections • i) First timers • ii) Subject Round • iii) Rapid fire round (Each team has to answer max. questions in 30 seconds.) • 10 marks for right answer and -5 for wrong answer. • The extra question will be asked in case there is tie between the teams.



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		● NOTE: if a college/institute has more than one team in the top 4, the The team having the top ranking will be selected for the final round.
33	One Minute JAM	● The number of participants is one. ● An on-the-spot topic will be given. ● Each participant will be given one minute to speak on a topic. ● Participants will be disqualified after exceeding the time limit. ● A participant has to speak for exactly one minute on the given topic without hesitation, repetition, or deviation. ● Judgment will be based on fluency, content, confidence, vocabulary, and time management. ● The decision of the judges will be final.
34	Youth Parliament	● Pacifism Vs Realism : The idea that all war is wrong versus the need to fight for the rights of a country. Is war justified? ● The number of participants per team is 1. ● A motion or bill will be given prior to the event. ● Participants will be assigned specific roles (speaker, ministers, opposition, etc.). ● Time limits for each speaker will be announced on the spot. ● Proper parliamentary etiquette, language, and decorum must be maintained. ● Judgment based on knowledge of the subject, oratory skills, logical arguments, body language, and adherence to rules. ● Use of abusive language will lead to disqualification. ● The decision of the judges will be final and binding.
35	Ad-Mad Show	● Each team should have 4-6 members only. ● The theme for the show will be imaginary product which is yet to be Launched. ● The time limit for performing the advertisement is 90-180 seconds. ● Participants shall be judged on the basis of spontaneity, content, adherence to the topic, presentation, coordination, and overall appeal of the advertisement. ● Caution should be taken to refrain from displaying obscenity, violence, prejudice, defamation, etc. in advertisements. ● Decisions of the judges will be final and binding.



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36	Treasure Hunt	• A team shall be of 3–4 members only. • Clues will be provided sequentially (tech-themed or related to the fest/college). • Total time allotted for the hunt will be announced at the beginning (approximately 60–90 minutes). • First team to solve all clues and reach the final destination will be the winner. • Use of unfair means, damaging property or misbehaviour will lead to disqualification. • Participants must follow all instructions of the coordinators strictly. • The decision of the organizers/judges will be final and binding.
37	OPTO (color blindness testup model)	• An apron & ID cards are a must. • Max. 5 members in one team. • Proper Explanation Should be within the time limit • Equipment will be provided on the spot.
38	Vital Monitoring	• An apron & ID cards are a must. • Max. 5 members in one team. • Proper Explanation Should be within the time limit • Equipment will be provided on the spot.
39	Cardiac Monitor	• An apron & ID cards are a must. • Max. 5 members in one team. • Proper Explanation Should be within the time limit • Equipment will be provided on the spot.
40	Hb Estimation & Introduction to different types of anemia	• An apron & ID cards are a must. • Max. 5 in one team. • Proper Explanation Should be within the time limit • Equipment will be provided on the spot.



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